

Lab 3 : Component Modelling and Architecture Pattern Selection

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1. Architectural Choice We selected the Layered (3-layer) architecture. The system is split into a Presentation Layer (User Interface Component, a web portal used by Trial Candidates and Clinical Researchers), a Business Layer (Recruitment Manager, Eligibility Screening, Consent Management, Visit Compliance and Audit Trail components) and a Data Layer (Trial Database Component). The Recruitment Manager coordinates everything, and the components talk only through interfaces: Recruitment, Eligibility, Consent, Visit, Audit and Database Interface.

2. Two Reasons for This Choice

- Each stage of the trial process has its own rules, so it fits into its own component. Eligibility screening, informed consent and visit compliance are three different jobs. Keeping them as separate components in the Business Layer means that if a trial protocol changes (for example a new exclusion criterion), we only update the Eligibility Screening Component and the criteria stored in the database. The UI and the other components stay untouched.
- Data must stay consistent and under central control. A candidate should never be marked as enrolled without a valid consent record, and visit logs must match the candidate's status. With one Recruitment Manager and one database this is simple to enforce with normal transactions. Microservices would spread this data across many services and make consistency harder, and a plain Client-Server setup gives no separation between screening,

consent and compliance logic. A layered system also has fewer moving parts, which makes it easier to validate for regulations like 21 CFR Part 11.

3. Security Advantage The web portal never accesses the database directly. Every request goes through the Recruitment Manager, which can check the user's role (candidate or researcher) before doing anything, so a candidate cannot view other candidates' data or edit trial criteria. Also, every important action (survey submission, consent, visit log) is sent to the Audit Trail Component, which stores tamper-evident entries (timestamped, signed and hash chained). This supports the audit trail requirement of 21 CFR Part 11.

4. Performance Benefit The layers are deployed together as one application, so calls between components are in-process calls without network delays between services. Eligibility screening is just a rule check of survey answers against stored criteria, so results come back quickly. The trial criteria are read often but change rarely, so they can be cached by the Recruitment Manager, which reduces repeated database queries when many candidates fill the survey at the same time.

Trade-off The whole application has to be scaled together instead of scaling one component alone, and the layers are somewhat tightly coupled. For a pre-screening platform of one research institution, this is acceptable compared to the simplicity, control and compliance benefits